ARC NetZero Hub End-of-Year Forum 2026

Monday 9 – Tuesday 10 November 2026 · Collaboration Lounge 4.10, Level 4, New Horizons (Building 82), 20 Research Way, Monash University Clayton

Day 1 — Monday 9 November

08:00–08:30 *Registration and coffee*

08:30–08:40 **Welcome and opening of the Forum**
Prof Wenhui Duan, Director, ARC NetZero Hub · Prof Ha Bui, Head, Department of Civil and Environmental Engineering, Monash University

08:40–10:25 **Academic session — research, national priorities, and the impact of AI on industry and research**
Chair: Prof Ranjith Pathegama Gamage, Monash University

- **Use of AI in geotechnical constitutive modelling** 25 min
  Prof John Carter, University of Newcastle
- **AI as a tool for engineers: from the rudimentary to the complex—limits, checks and balances** 25 min
  Prof Stephen Foster, UNSW Sydney
- **Building to last: reducing carbon footprint through durability** 25 min
  A/Prof Sze Dai Pang, National University of Singapore
- **The challenges for Australia offshore wind and potential foundation solutions** 15 min
  Prof Yinghui Tian, University of Melbourne
- **Beyond automation: physical human–robot collaboration and the future of intelligent engineering** 15 min
  Prof Ying Tan, University of Melbourne

10:25–10:45 *Group photograph, then morning tea*

10:45–12:30 **Industry session — challenges on the way to net zero**
Chair: Prof Malek Bouazza, Monash University

- **Industry challenges in transport practice** 25 min
  Dr Richard Yeo, National Transport Research Organisation (NTRO)
- **Landfill lifecycle and greenhouse gas** 25 min
  Dr Asgar Ali, Wyndham City Council
- **Proposal: Climate-Resilient Infrastructure and Modular Building (CLIMB) Hub** 15 min
  Prof Wei Gao, UNSW Sydney
- **Proposal: Adaptive and Sustainable Infrastructure (ASI) Hub** 15 min
  A/Prof Xihong Zhang, Curtin University
- **Industry challenges on carbon emissions** 25 min
  Dr Scott Taylor, VIDA Roads

12:30–13:10 *Lunch*

12:35–13:05 **Lunchtime online talk**
Host: Prof Michael Kennedy OAM, Director, Monash Institute of Transport Studies

- **A marathon career** 30 min
  Jessica Stenson, three-time Olympian and Commonwealth Games marathon champion (online)

13:10–15:15 **Women Leaders in Engineering**

Chair: Dr Mark Toner, Toner & Associates

- **Welcome and framing by the chair** 5 min
- **Keynote — carbon emissions and the energy transition** 30 min
  Leeanne Bond FTSE FAICD, Vice-President (Membership), ATSE
- **Material matters: engineering decisions, net zero and women leading change** 15 min
  Prof Hong Guan, Griffith University
- **Leading with purpose: driving sustainable transformation in concrete infrastructure** 15 min
  Prof Rebecca Gravina, University of Queensland
- **A journey towards a NetZero built environment** 15 min
  A/Prof Hong Xian Li, Deakin University
- **AI for fire safety engineering** 15 min
  Prof Kate Nguyen, RMIT University
- **Beyond the nozzle: process-induced microstructure and anisotropy in 3D concrete printing** 15 min
  A/Prof Xuemei Liu, University of Melbourne
- **TREMS – next stage: industrial transformation across sectors** 15 min
  Distinguished Prof Sujeeva Setunge, RMIT University

15:15–15:25 *Afternoon tea*

15:25–16:40 **ARC research infrastructure session**

Chair: A/Prof Hossein Masoumi, Monash University

- **Collaborative robotics for structural assembly (LE210100019) — capability, access and first results** 25 min
  A/Prof Yihai Fang, Monash University
- **Ultrafast dynamic tomography and X-ray based rheography facility — TESCAN DynaTOM (LE250100091)** 25 min
  A/Prof Mohammad Saadatfar, University of Sydney
- **Experimental research of heavy rail systems** 25 min
  A/Prof Qing Wu, Centre for Railway Engineering, CQUniversity

16:40–17:30 **Launch of the NetZero and NØBEL mentoring programme**

Chair: Prof Hao Wang, Swinburne University of Technology

- **Opening** 10 min
  Prof Wenhui Duan · Distinguished Prof Vivian Tam, ARC ITTC NØBEL Centre
- **The mentoring programme — format, pairing and sign-up** 10 min
  Dr Felipe Basquiroto de Souza, Hub Manager · Anush Heinitz, NØBEL Centre Manager
- **Mentoring experience** 10 min
  Dr Chamila Gunasekara, RMIT University
- **Mentees' perspective** 20 min
  Kun Chen, RMIT University · Amir Mohamadi, Monash University · Goran Shirzad, Monash University

17:30–18:00 *Reception · Monash Club*

18:00–20:00 **Hub dinner · Monash Club**

MC: Prof Priyan Mendis, University of Melbourne

- **18:30–18:35 · Dinner address and introduction of the ATSE speaker** 5 min
  Prof Yiannis Ventikos, Dean of Engineering, Monash University
- **18:35–19:15 · The ATSE nomination process: how new Fellows are elected** 40 min
  Leeanne Bond FTSE FAICD, Vice-President (Membership), ATSE

Day 2 — Tuesday 10 November · Hub programme

08:00–08:45 *RTT Committee (members only) · Room 4.07/4.08 · Chair: Prof Lihai Zhang*

08:30–08:45 *Arrival and coffee*

08:45–09:50 **Academic session**

Chair: Prof Rebecca Gravina, University of Queensland

- **Revisiting taken-for-granted issues on toughening of polymer nanocomposites** 25 min
  Prof Yiu-Wing Mai, The Hong Kong Polytechnic University · Chair, Hub Advisory Committee
- **Physical human–robot collaboration in the AI era** 25 min
  Prof Dikai Liu, University of Technology Sydney
- **Recent advances in corrosion science** 15 min
  A/Prof Sebastian Thomas, Monash University

09:50–11:00 **Hub project presentations I**

Chair: Prof Qing Li, The University of Sydney

- **Connected data, agentic AI and digital-twin carbon accounting (IH24.18, IH24.19)** 20 min
  Prof Peng Wu, Curtin University · Dr Jun Wang, Western Sydney University
- **Development of low-carbon technologies in concrete and ground improvement (IH24.05, IH24.15)** 20 min
  Dr Thanh Nguyen, University of Technology Sydney
- **Digital twin for a production line (IH24.07)** 10 min
  Dr Yunlong Tang, Monash University
- **Advanced deep learning approaches for image-driven soil classification and property prediction (IH24.04)** 10 min
  Prof Dilan Robert, RMIT University
- **LC3 meets CCUS (IH24.12)** 10 min
  Dr Cheng Jiang, Western Sydney University

11:00–11:10 *Morning tea*

11:10–12:30 **Hub project presentations II**

Chair: Prof Dilan Robert, RMIT University

- **Road profiling, AI assistants and robotic survey in mining (IH24.06, IH24.10, IH24.20)** 20 min
  Prof Chao Chen, Monash University
- **Characterisation of cemented paste backfill (CPB) and paste waste behaviour used in underground and surface mining spaces (IH24.13, IH24.16)** 20 min
  Amir Mohamadi, Monash University (for A/Prof Hossein Masoumi)
- **Sustainable façade system for improved thermal performance (IH24.11, IH24.17)** 20 min
  Prof Pat Rajeev, Swinburne University of Technology
- **AI-enabled service-oriented digital twin framework (IH24.01)** 10 min
  Prof Lihai Zhang, University of Melbourne
- **Waste-based cementitious materials and aggregates (IH24.08)** 10 min
  Prof Zhong Tao, Western Sydney University

12:30–13:05 **Hub performance and future — the ARC Prioritise proposal**

Chair: Prof Lihai Zhang, University of Melbourne

- **The Hub in 2026 — overview, finances and KPIs; the proposed ARC Prioritise Centre for Planetary Robotics** 35 min
  Prof Wenhui Duan, Director, ARC NetZero Hub

13:05–14:00 *Lunch and networking · Advisory Committee meeting (members only), Room 4.07/4.08, 13:10–14:10*

14:00 *Forum closes*

Each presentation slot includes questions. ARC Research Hub for Infrastructure Net Zero (IH240100006). Programme subject to change.